

Tiago Garrel

AI & Data Engineer · MSc in Big Data & Analytics

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PROFESSIONAL SUMMARY

Production Engineer with a Master's in Big Data & Analytics, combining a strong foundation in operations, process optimization, and data engineering with growing hands-on experience in AI and automation. Over two years applying Python, SQL, and BI tools in real supply chain and analytics environments; **currently building agentic systems and LLM-powered pipelines as part of my MSc thesis and personal projects**. From ETL design and cloud infrastructure to API development and ML models. Native Spanish speaker, C2 English (Cambridge CPE).

TECHNICAL SKILLS

- **Languages & Libraries:** Python (asyncio, Pydantic, Pandas, Polars, PyArrow) · SQL
- **Data Engineering:** ETL/ELT pipeline design · Medallion Architecture · Apache Airflow · Parquet · data modelling · idempotent pipelines
- **Cloud & Infrastructure:** AWS S3 · Docker · Docker Compose · BigQuery · boto3 · Git
- **AI & LLM:** OpenRouter · OpenSource LLMs (llama.cpp / GGUF) · multi-agent orchestration · RAG (HyDE, MMR) · FAISS · FastAPI
- **ML:** Random Forest · LightGBM · predictive modelling · mathematical optimisation
- **BI & Analytics:** Power BI · Looker · Google Sheets · dashboard development · competitive intelligence
- **APIs & Integration:** REST API design · FastAPI · SSE streaming · cross-system integrations

PROFESSIONAL EXPERIENCE

Independent Consultant · Natural Cosmetics Brand · Remote March 2026 – Present

- Designing and building end-to-end operational infrastructure from scratch: database architecture, stock management logic, automated production systems, and cross-system integrations. Pending full evaluation on improvement but already reducing over 1h a day in operations manual work.

Competition Insights Analyst · eDreams ODIGEO · Barcelona, Spain November 2025 – February 2026

- Built Python and Google Apps Script-based tools that automated recurring analytical workflows, reducing 15 to 20 minutes in manual tasks daily.
- Developed a system to detect and track anomalies, reducing detection time from 2.5 weeks on average to 2 days.

Data & Operations Analyst · Ártico · Canelones, Uruguay November 2022 – June 2024

- Successfully led a company-wide implementation of a TMS software, coordinating cross-departmental rollout and ensuring on-time delivery. Assumed the project manager role with the project delayed by 2 months; over 20 people were directly implicated with the tool.
- Developed algorithms and management tools for automated warehouse slotting optimisation across 1,500+ SKUs, improving operations by over 20%, reducing considerably warehouse movements.

Supervisor · Mayohogar · Montevideo, Uruguay February 2022 – October 2022

- Managed personnel (3) and operations, improving revenue 70% in MoM in 6 months.

Co-founder · UnMonton Board Games · Uruguay 2020 – 2021

- Co-developed and launched a board-game brand: game design, prototyping, production, and marketing; sustained for one year through effective community engagement. Sold over 200 games nationwide.

TECHNICAL PROJECTS

Conversational Analytics System — Master's Thesis (2026)

- End-to-end agentic analytics platform for insurance call centre: Data pipeline with Medallion Architecture on AWS S3 with LLM-enrichment, evaluator-retry for quality control, FAISS semantic search, and a multi-agent system with RAG using HyDE + MMR.
- **Stack:** Python · Gemma 3 (llama.cpp/GGUF) · OpenRouter · FAISS · FastAPI · AWS S3 · Docker Compose.

[AI & Technology Blog · tiagogarrel.github.io](https://tiagogarrel.github.io)

- Blog and content platform covering AI systems, data engineering, and emerging technology for technical and non-technical audiences. 5+ published posts across LinkedIn and the personal blog.

CV Tuning & Application Tracking Platform · (In Development)

- AI-powered platform to personalise CVs for specific job postings and track the full job application pipeline.

Bachelor's Thesis: Route Optimisation in Frozen Goods Distribution

- Mathematical model to minimise distribution costs under complex constraints; outperformed commercial software, was awarded a 100% grade, and was presented at the XXII South American Conference of Industrial, Production & Mechanical Engineering.

EDUCATION

Master's in Big Data & Analytics · EAE Business School · Barcelona, Spain October 2025 – July 2026

Bachelor's Degree in Production Engineering · UdelaR · Montevideo, Uruguay 2019 – 2023

CERTIFICATIONS

- Certificate of Proficiency in English (C2), Cambridge University
- Agent Evaluation on Databricks
- Data Modelling, Transformation & Serving — DeepLearning.AI (Coursera)
- IBM Project Manager Professional Certificate
- Python for Data Science & Machine Learning Bootcamp — Udemy
- Microsoft Power BI Desktop for Business Intelligence — Udemy
- The Complete SQL Bootcamp 2022: Go from Zero to Hero — Udemy